

An AI-assisted note on the dp-rank of ordered abelian groups

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Introduction

This note presents a proof of the known computation of the dp-rank of ordered abelian groups in the pure language of ordered groups. Gurevich and Schmitt showed that every ordered abelian group has NIP [5], and hence has dp-rank at most $\aleph_0$. For background on NIP and dp-rank, we refer to Simon's guide [8]. If $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank is $\aleph_0$. If instead G is nontrivial and has finite spines, then

$$\text{dp-rk}(G) = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Here $\mathbb{P}$ is the set of primes, and $\mathcal{S}_p^\infty$ is the set of nonempty $H \in \mathcal{S}_p$ for which the rib $R_H^{s_p}$ is infinite. None of these results is new: the formula is due to Farré [4], while related characterizations of ordered abelian groups of finite dp-rank were obtained independently by Dolich and Goodrick [2] and by Halevi and Hasson [6].

The purpose of this note is to practice AI-assisted proof writing. The author supplied sketches of the proofs; ChatGPT 5.6 Sol and Grok 4.6 expanded them into complete proofs, which the author subsequently reviewed and revised. For background on ordered abelian groups, we refer to Cluckers and Halupczok [1]. The notation follows [7], which is itself based on their framework.

1 The infinite spine case

Throughout this note, G denotes an ordered abelian group. For $n \geq 2$ and $a \notin nG$, recall that $\mathfrak{s}_n(a)$ is the largest convex subgroup $H \subseteq G$ such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$, and we regard $\emptyset$ as being smaller than every convex subgroup. We set $\mathcal{S}_n = \{\mathfrak{s}_n(a) \mid a \in G\}$. The subgroups $\mathfrak{s}_n(a)$ are uniformly definable in a [1, Lemma 2.1]. Consequently, comparisons and equalities between values of the maps $\mathfrak{s}_n$ are formulas in L_{oag} .

Lemma 1.1 ([7, Lemma 2.8(2) and Remark 2.6]). *Let p be a prime, $r \geq 1$, and $a, b \in G$.*

1. *If $0 \leq j < r$, then*

$$\mathfrak{s}_{p^r}(p^j a) = \mathfrak{s}_{p^{r-j}}(a).$$

If $j \geq r$, then $\mathfrak{s}_{p^r}(p^j a) = \emptyset$.

2. *If $\mathfrak{s}_{p^r}(a) \neq \mathfrak{s}_{p^r}(b)$, then*

$$\mathfrak{s}_{p^r}(a + b) = \max\{\mathfrak{s}_{p^r}(a), \mathfrak{s}_{p^r}(b)\}.$$

Lemma 1.2. *Let p be a prime and let $H < H'$ be nonempty elements of $\mathcal{S}_p$. There exists $a \in G$ such that*

$$\mathfrak{s}_p(a) = H \quad \text{and} \quad \mathfrak{s}_{p^r}(a) < H'$$

for every $r \geq 1$.

Proof. Choose $b \in G$ such that $\mathfrak{s}_p(b) = H$. Since $H' > H$, the maximality of H gives $b \in H' + pG$. We may therefore write $b = a + pc$ with $a \in H'$ and $c \in G$. As $a - b \in pG$, we have $\mathfrak{s}_p(a) = \mathfrak{s}_p(b) = H$. If $\mathfrak{s}_{p^r}(a) \geq H'$ for some $r \geq 1$, then $a \in H' \subseteq \mathfrak{s}_{p^r}(a)$, contrary to $a \notin \mathfrak{s}_{p^r}(a) + p^r G$. Therefore $\mathfrak{s}_{p^r}(a) < H'$. ■

Lemma 1.3 (Finite coding). *Let p be a prime and*

$$K_0 < K_1 < \cdots < K_N$$

be nonempty elements of $\mathcal{S}_p$. There exists $x \in G$ such that

$$\mathfrak{s}_{p^{i+1}}(x) = K_i$$

for every $0 \leq i \leq N$.

Proof. For each $j < N$, use Lemma 1.2 to choose $a_j \in G$ such that

$$\mathfrak{s}_p(a_j) = K_j \quad \text{and} \quad \mathfrak{s}_{p^r}(a_j) < K_{j+1}$$

for every $r \geq 1$. Also choose $a_N \in G$ with $\mathfrak{s}_p(a_N) = K_N$, and set

$$x = a_0 + pa_1 + \cdots + p^N a_N.$$

Fix $0 \leq i \leq N$. By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i) = \mathfrak{s}_p(a_i) = K_i.$$

If $j < i$, then

$$\mathfrak{s}_{p^{i+1}}(p^j a_j) = \mathfrak{s}_{p^{i+1-j}}(a_j) < K_{j+1} \leq K_i,$$

while, if $j > i$, then $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \emptyset$. Thus, among the summands of x , the unique maximal $\mathfrak{s}_{p^{i+1}}$ -value is K_i . Repeatedly applying Lemma 1.1(2) gives $\mathfrak{s}_{p^{i+1}}(x) = K_i$. ■

Theorem 1.4. *If $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank of G is $\aleph_0$.*

Proof. We work in a sufficiently saturated monster model of $\text{Th}(G)$. Since $\mathcal{S}_p$ is an infinite interpretable linear order, compactness allows us to choose elements

$$(b_{i,k})_{i,k < \omega}$$

such that all the values

$$H_{i,k} := \mathfrak{s}_p(b_{i,k})$$

are nonempty and

$$H_{i,k} < H_{j,l} \quad \text{whenever} \quad i < j \text{ or } (i = j \text{ and } k < l).$$

For a function $\eta : \omega \rightarrow \omega$, consider the partial type

$$\Sigma_\eta(x) = \{ \mathfrak{s}_{p^{i+1}}(x) = H_{i,\eta(i)} \mid i < \omega \}.$$

This type is finitely satisfiable. Indeed, given $N < \omega$, the values

$$H_{0,\eta(0)} < H_{1,\eta(1)} < \cdots < H_{N,\eta(N)}$$

form a finite increasing chain, so Lemma 1.3 realizes the first $N + 1$ conditions of Σ_η . By saturation, there is $a_\eta \in G$ realizing Σ_η .

For every $i < \omega$, let

$$\varphi_i(x; y) \quad \text{be the formula} \quad \mathfrak{s}_{p^{i+1}}(x) = \mathfrak{s}_p(y).$$

As noted above, this is an L_{oag} -formula. For all $\eta : \omega \rightarrow \omega$, $i < \omega$, and $k < \omega$, we have

$$\models \varphi_i(a_\eta; b_{i,k}) \iff H_{i,\eta(i)} = H_{i,k} \iff \eta(i) = k.$$

Hence, these formulas and parameters form an ICT-pattern of depth $\aleph_0$, and therefore the dp-rank of G is at least $\aleph_0$.

Every ordered abelian group has NIP [5]. Since L_{oag} is countable, this gives dp-rank at most $\aleph_0$. Consequently, the dp-rank of G is exactly $\aleph_0$. ■

2 The lower bound in the finite-spine case

We now assume that G is nontrivial and has finite spines: $\mathcal{S}_p$ is finite for every prime p . For a prime p and $H \in \mathcal{S}_p \setminus \{\emptyset\}$, recall that

$$H^{[p]} = \bigcap_{K \in G, K > H} (K + pG)$$

and

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

by [7, Definition 2.10 and Lemma 2.14(1)]. We also use $\mathcal{S}_{p^r} = \mathcal{S}_p$ for every $r \geq 1$, as established in [1, Lemma 2.2(1)]. We put

$$\mathcal{S}_p^\infty = \{H \in \mathcal{S}_p \setminus \{\emptyset\} \mid |R_H^{\mathfrak{s}_p}| = \infty\}.$$

Lemma 2.1. *Let p be a prime and $H \in \mathcal{S}_p^\infty$. There is a sequence $(a^k)_{k < \omega}$ such that*

$$\mathfrak{s}_p(a^k) = \mathfrak{s}_p(a^k - a^l) = H$$

whenever $k \neq l$. If $H < K$ for some $K \in \mathcal{S}_p \setminus \{\emptyset\}$, the sequence may moreover be chosen inside K .

Proof. Choose elements $(u^k)_{k < \omega}$ whose images in

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

are nonzero and pairwise distinct. Thus, neither u^k nor $u^k - u^l$ belongs to $H + pG$, while all these elements belong to $H^{[p]}$. By [7, Lemma 2.14(1)], this says precisely that

$$\mathfrak{s}_p(u^k) = \mathfrak{s}_p(u^k - u^l) = H$$

for $k \neq l$.

Suppose that $H < K$. Since $u^k \in H^{[p]} \subseteq K + pG$, we can write $u^k = a^k + pc_k$ with $a^k \in K$. Replacing u^k by a^k does not change its image in $R_H^{\mathfrak{s}_p}$. The images therefore remain nonzero and pairwise distinct, so the required equalities remain true. $\blacksquare$

Lemma 2.2 (Simultaneous coding for one prime). *Let p be a prime and let*

$$H_0 < H_1 < \cdots < H_{n-1}$$

be elements of $\mathcal{S}_p^\infty$. For each $i < n$, there is a sequence $(a_i^k)_{k < \omega}$ as in Lemma 2.1 for H_i , and these sequences can be chosen so that $a_i^k \in H_{i+1}$ whenever $i < n-1$. For $\zeta : n \rightarrow \omega$, set

$$b_\zeta = a_0^{\zeta(0)} + pa_1^{\zeta(1)} + \cdots + p^{n-1}a_{n-1}^{\zeta(n-1)}.$$

Then, for every $i < n$ and every $k < \omega$,

$$b_\zeta - p^i a_i^k \in H_i + p^{i+1}G \iff k = \zeta(i).$$

Proof. The required sequences are supplied by Lemma 2.1. For $j < i$, we have $a_j^{\zeta(j)} \in H_{j+1} \subseteq H_i$, whereas for $j > i$, the term $p^j a_j^{\zeta(j)}$ belongs to $p^{i+1}G$. Hence,

$$b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G.$$

If $k \neq \zeta(i)$, then $\mathfrak{s}_{p^{i+1}}(p^i(a_i^{\zeta(i)} - a_i^k)) = \mathfrak{s}_p(a_i^{\zeta(i)} - a_i^k) = H_i$, so

$$p^i a_i^{\zeta(i)} - p^i a_i^k \notin H_i + p^{i+1}G.$$

Together with the preceding congruence, this proves the claim. $\blacksquare$

The following standard consequences of compactness will be used without proof. For the second assertion, finite satisfiability follows from the Chinese remainder theorem [7, Lemma 2.1(2)].

Lemma 2.3. *Let G be a sufficiently saturated monster model of the theory of a nontrivial ordered abelian group.*

1. *The subgroup $\bigcap_{m \geq 2} mG$ is cofinal and coinital in G .*
2. *Let $P_0 \subseteq \mathbb{P}$ and $(a_p)_{p \in P_0}$ be a sequence from G . There exists $a \in G$ such that*

$$a - a_p \in p^r G$$

for every $p \in P_0$ and $r \geq 1$.

Theorem 2.4. *Let G be a nontrivial ordered abelian group with finite spines. Then*

$$\text{dp-rk}(G) \geq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Proof. We work in a sufficiently saturated monster model. For each prime p , put $n_p = |\mathcal{S}_p^\infty|$ and, when $n_p > 0$, enumerate

$$\mathcal{S}_p^\infty = \{H_{p,0} < \cdots < H_{p,n_p-1}\}.$$

The number n_p is finite because G has finite spines. Let

$$\mathcal{I} = \{(p, i) \mid p \in \mathbb{P}, 0 \leq i < n_p\}.$$

Thus,

$$|\mathcal{I}| = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

and $\mathcal{I}$ is finite or countable.

For each p with $n_p > 0$, choose sequences $(a_{p,i}^k)_{k < \omega}$ as in Lemma 2.2, and denote the corresponding coding elements by $b_{p,\zeta}$ for $\zeta : n_p \rightarrow \omega$. For $(p, i) \in \mathcal{I}$, let

$$\varphi_{p,i}(x; y) \quad \text{be the formula} \quad x - y \in \mathfrak{s}_{p^{i+1}}(y) + p^{i+1}G.$$

This is an L_{oag} -formula by the uniform definability of $\mathfrak{s}_{p^{i+1}}$. Lemma 1.1 implies $\mathfrak{s}_{p^{i+1}}(p^i a_{p,i}^k) = \mathfrak{s}_p(a_{p,i}^k) = H_{p,i}$ for every $k < \omega$, and hence

$$\models \varphi_{p,i}(b_{p,\zeta}; p^i a_{p,i}^k) \iff k = \zeta(i).$$

Fix a function $\eta : \mathcal{I} \rightarrow \omega$. For each p with $n_p > 0$, let

$$\eta_p : n_p \rightarrow \omega, \quad \eta_p(i) = \eta(p, i).$$

By Lemma 2.3(2), there is $b_\eta \in G$ such that

$$b_\eta - b_{p,\eta_p} \in p^{i+1}G$$

for every p with $n_p > 0$ and $i \geq 0$. Since translation by an element of $p^{i+1}G$ does not change the truth of formula $\varphi_{p,i}$, we have

$$\models \varphi_{p,i}(b_\eta; p^i a_{p,i}^k) \iff k = \eta(p, i)$$

for all $(p, i) \in \mathcal{I}$ and $k < \omega$.

It remains to add the row arising from the order. Since the monster is sufficiently saturated, there exists $c > 0$ such that

$$-c < b_\eta < c$$

for every $\eta : \mathcal{I} \rightarrow \omega$. By Lemma 2.3(1), choose

$$t_0 < t_1 < \dots$$

from $\bigcap_{m \geq 2} mG$ so that $t_{k+1} - t_k > 2c$ for every $k < \omega$. Set

$$I_k = (t_k - c, t_k + c).$$

These intervals are pairwise disjoint and satisfy $I_0 < I_1 < \dots$. We take

$$\psi(x; y, z) := (y < x < z)$$

for the additional row, with parameter sequence

$$(t_k - c, t_k + c)_{k < \omega}.$$

Let $\nu : \{*\} \cup \mathcal{I} \rightarrow \omega$, and put $\eta = \nu|_{\mathcal{I}}$. The element

$$a_\nu = b_\eta + t_{\nu(*)}$$

belongs to $I_{\nu(*)}$. Moreover, $t_{\nu(*)} \in p^{i+1}G$ for every $(p, i) \in \mathcal{I}$, so translation by $t_{\nu(*)}$ does not change the truth of any formula $\varphi_{p,i}$. It follows that

$$\models \psi(a_\nu; t_k - c, t_k + c) \iff \nu(*) = k$$

and

$$\models \varphi_{p,i}(a_\nu; p^i a_{p,i}^k) \iff \nu(p, i) = k.$$

We have therefore constructed an ICT-pattern of depth

$$1 + |\mathcal{I}| = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

as required. ■

3 The upper bound in the finite-spine case

We continue to assume that G is nontrivial and has finite spines, and now also assume that

$$N = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. We use the following standard criterion.

Fact 3.1 (Indiscernible-sequence criterion). *Let T be a complete theory, let $n \geq 1$. Fix a Dedekind complete $I \models DLO$. Suppose that, whenever $(\bar{a}_i)_{i \in I}$ is an indiscernible sequence and b is a singleton, one can divide I by at most n points so that*

$$\text{tp}(\bar{a}_i/b) = \text{tp}(\bar{a}_j/b)$$

whenever i and j lie in the same convex component of the remainder. Then $\text{dp-rk}(T) \leq n$.

This may be proved in the same way as [3, Fact 2.10]. Henceforth I denotes this Dedekind complete index set. For $p \in \mathbb{P}$ and $a \in G$, recall that $\mathfrak{t}_p(a) = \bigcup_{H \in \mathcal{S}_p, a \notin H} H$. We set $\mathcal{T}_p = \{\mathfrak{t}_p(a) \mid a \in G\}$, but this coincides with $\mathcal{S}_p$ if G has finite spines. We will also use the following result originating in the relative quantifier elimination by Cluckers and Halupczok [1]. See [7, Definition 2.20] for the notation.

Proposition 3.2. *Let G be an ordered abelian group with finite spines and let f be a partial map from G to G whose domain is a subgroup. Suppose that f is an ordered group embedding and, for every a in its domain, every prime p , and every $r \geq 1$, the following hold:*

$$\begin{aligned} \mathfrak{s}_{p^r}(f(a)) &= \mathfrak{s}_{p^r}(a), & \mathfrak{t}_p(f(a)) &= \mathfrak{t}_p(a), \\ a = \bullet \cdot k_\bullet &\iff f(a) = \bullet \cdot k_\bullet, & a \equiv_{p^r, \bullet} k_\bullet &\iff f(a) \equiv_{p^r, \bullet} k_\bullet. \end{aligned}$$

Here the third condition is required for every $k \in \mathbb{Z} \setminus \{0\}$, and the fourth for every $1 \leq k < p^r$. Then f is elementary.

Proof. This is a special case of [7, Proposition 2.26]. We use the identity map on all the auxiliary sorts in the cited proposition. We explain why predicates $D_{p^r}^{[p^s]}$, where $s \geq r$, require no additional hypothesis. Put $H = \mathfrak{s}_{p^r}(a)$. If $H = \emptyset$ then $D_{p^r}^{[p^s]}(a)$ does not hold. Suppose that $H \neq \emptyset$. If H has a successor H' in $\mathcal{S}_p$ then $H^{[p^s]} = H' + p^s G$, while if H has no successor then $H^{[p^s]} = G$. In either case, it automatically follows that $a \in H^{[p^s]} + p^r G$, which implies $D_{p^r}^{[p^s]}(a)$. This completes the proof. $\blacksquare$

Lemma 3.3 ([7, Lemma 3.4]). *Let p be a prime, $H \in \mathcal{S}_p \setminus \{\emptyset\}$, and $r \geq 1$. If $R_H^{\mathfrak{s}_p}$ is finite, then $R_H^{\mathfrak{s}_{p^r}}$ is finite. Consequently,*

$$|R_H^{\mathfrak{s}_{p^r}}| = \infty \implies H \in \mathcal{S}_p^\infty.$$

The next lemma isolates the single contribution coming from the order and from the ordered ribs associated with the maps $\mathbf{t}_p$.

Lemma 3.4 (The common order cut). *Let $(\bar{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. One can divide I by at most one point so that the following data are constant on each convex component of the remainder:*

1. the sign of $\bar{m} \cdot \bar{a}_i - mb$;
2. $\mathbf{t}_p(\bar{m} \cdot \bar{a}_i - mb)$;
3. the truth value of $\bar{m} \cdot \bar{a}_i - mb =_{\bullet} k_{\bullet}$.

Here the assertions hold simultaneously for all integer tuples $\bar{m}$, all nonzero integers m , all primes p , and all nonzero integers k .

Proof. Let $\mathcal{H}$ consist of $\{0\}$ and all nonempty elements of the finite sorts $\mathcal{T}_p$. Finiteness of every $\mathcal{T}_p$ implies that an indiscernible sequence remains indiscernible with respect to formulas involving elements of $\mathcal{H}$.

Fix $H \in \mathcal{H}$, an integer tuple $\bar{m}$, a nonzero integer m , and an integer k , where either $k = 0$ or G/H is discrete. Let

$$c_i = (\bar{m} \cdot \bar{a}_i)/H - k_H \in G/H.$$

$(c_i)_{i \in I}$ is also indiscernible. In particular, it is either constant or strictly monotone. Consequently, exactly one of the following holds:

- (i) the sign of $c_i - m(b/H)$ is constant as i varies in I ;
- (ii) there exists $i_0 \in I$ such that the terms $c_i - m(b/H)$ for $i < i_0$ are all negative and those terms for $i > i_0$ are all positive, or vice versa.

(The realization of the cut as a point of I uses Dedekind completeness.) Now suppose that (ii) holds with cutting point i_0 , and let $K \in \mathcal{H}$ satisfy $K > H$. Then $k_H/K = 0$ and hence $c_i/K = (\bar{m} \cdot \bar{a}_i)/K$. Thus, either (i) or (ii) holds also for $c_i/K - m(b/K)$. If (ii) holds, then the cutting point is the same i_0 because $c_i/K - m(b/K)$ for $i < i_0$ are nonpositive and those for $i > i_0$ are nonnegative, or vice versa.

We next show that two distinct points of I cannot arise as cutting points in (ii). Let $c_i = \bar{m} \cdot \bar{a}_i/H - k_H$ and $d_i = \bar{m}' \cdot \bar{a}_i/K - k'_K$ with $H \leq K$. Suppose that distinct i_0 and j_0 in I arise as cutting points of $c_i - m(b/H)$ and $d_i - m'(b/K)$ in (ii), respectively. We may assume that $i_0 < j_0$. We also assume that $c_i - m(b/H)$ for $i < i_0$ are negative by negating the sequence if necessary. Furthermore, we may assume that $m = m'$ by multiplying both sequences by some nonzero integers if necessary. Choose

$$h < i_0 < r < s < j_0 < l.$$

If (i) holds for $c_i/K - m(b/K)$, then $c_i/K = m(b/K)$ for all i . Hence the signs of $d_s - c_{i_0}/K$ and $d_l - c_{i_0}/K$ are different, contradicting the indiscernibility

of $(c_i/K, d_i)_{i \in I}$. Therefore we assume that (ii) holds for $c_i/K - m(b/K)$ with cutting point i_0 , and denote c_i/K by c_i henceforth. If (d_i) is also increasing, then we have $c_r > d_s$ but $c_h < d_l$, contradicting indiscernibility. If instead (d_i) is decreasing, then $c_h < d_h$ but $c_l > d_l$, again leading to a contradiction.

Thus at most one cutting point of I arises. Dividing I by this point if it exists, the signs under consideration are constant on each convex component of the remainder. Taking $H = \{0\}$ and $k = 0$ gives (1). (2) follows because $\mathfrak{t}_p(x) < H$ if and only if $x \in H$ if and only if $x/H = 0$ for $H \in \mathcal{T}_p$. Finally, by [7, Lemma 2.21(1)], a witness to $x = \bullet k_\bullet$ is $\mathfrak{t}_2(x)$. Thus (2), together with constancy of the sign of $x/H - k_H$, gives (3). $\blacksquare$

We now control all contributions from the $\mathfrak{s}_{p^r}$ -side. Let $(\bar{a}_i)_{i \in I}$ and b be as in Lemma 3.4. For a prime p and for $H \in \mathcal{S}_p \setminus \{\emptyset\}$, call $i_0 \in I$ a (p, H) -exception if there are an integer tuple $\bar{m}$, a nonzero integer m , an integer $r \geq 1$, and an integer k such that either $k = 0$ or G/H is discrete, and such that, writing

$$c_i = (\bar{m} \cdot \bar{a}_i)/H - k_H \in G/H,$$

one has

$$c_i \equiv_{p^r} m(b/H) \iff i = i_0.$$

Lemma 3.5 (Uniqueness of the congruence exception). *Let p be a prime and let $H \in \mathcal{S}_p \setminus \{\emptyset\}$.*

1. *There is at most one (p, H) -exception.*
2. *If i_0 is a (p, H) -exception, then i_0 is also a (p, K) -exception for some $K \in \mathcal{S}_p^\infty$.*

Proof. For (1), suppose that $i_0 < j_0$ are (p, H) -exceptions, witnessed respectively by

$$(c_i, m, r) \quad \text{and} \quad (d_i, m', r')$$

in the notation above. In a torsion-free abelian group, one has $x \equiv_{p^s} 0 \iff p^t x \equiv_{p^{s+t}} 0$ for all $s, t \geq 1$. It allows us to assume that $r = r'$. One also has $x \equiv_{p^s} 0 \iff ux \equiv_{p^s} 0$ whenever u is coprime to p . By writing $m = np^e$ where n is coprime to p and multiplying the inverse of n in $\mathbb{Z}/p^r\mathbb{Z}$, we also may assume that $m = p^e$, and similarly that $m' = p^f$. We now have

$$\begin{aligned} c_i \equiv_{p^r} p^e(b/H) &\iff i = i_0, \\ d_i \equiv_{p^r} p^f(b/H) &\iff i = j_0. \end{aligned}$$

If $e \leq f$, eliminating b gives

$$d_{j_0} - p^{f-e} c_{i_0} \equiv_{p^r} 0.$$

By indiscernibility, the same relation holds with j_0 replaced by any $j > i_0$. Subtracting yields $d_j \equiv_{p^r} d_{j_0}$, so the second exceptional congruence also holds at j , a contradiction. The case $e > f$ is symmetric.

For (2), suppose again (c_i, m, r) witnesses that i_0 is a (p, H) -exception, where $c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H$. By indiscernibility, the value of $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_j - \overline{m} \cdot \overline{a}_i)$ is constant when distinct i, j range over I . Let K be this value. We have $K \geq H$ because if otherwise, $c_i \equiv_{p^r} c_j$ for all distinct i, j , which contradicts the definition of i_0 . It follows that $c_{i_0}/K \equiv_{p^r} m(b/K)$. $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - \overline{m} \cdot \overline{a}_{i_0}) = K$ implies $c_i/K \not\equiv_{p^r} c_{i_0}/K$ for $i \neq i_0$, and thus $c_i/K \not\equiv_{p^r} m(b/K)$, which shows that i_0 is also a (p, K) -exception. The images of $(\overline{m} \cdot \overline{a}_i - \overline{m} \cdot \overline{a}_{i_0})_{i \neq i_0}$ in $R_K^{\mathfrak{s}_{p^r}}$ are pairwise distinct, hence $K \in \mathcal{S}_p^\infty$ by Lemma 3.3. $\blacksquare$

Lemma 3.6 (Control of the $\mathfrak{s}$ -data). *For every pair (p, H) with $H \in \mathcal{S}_p^\infty$, choose its exceptional index if one exists, and let E be the set of indices so obtained. Then $|E| \leq N$. If $C \subseteq I$ is convex and $C \cap E = \emptyset$, then for any $i, j \in C$, every integer tuple $\overline{m}$, every nonzero integer m , every prime p , every $r \geq 1$, and every $1 \leq k < p^r$, we have*

$$\begin{aligned} \mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - mb) &= \mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_j - mb), \\ \overline{m} \cdot \overline{a}_i - mb \equiv_{p^r, \bullet} k_\bullet &\iff \overline{m} \cdot \overline{a}_j - mb \equiv_{p^r, \bullet} k_\bullet. \end{aligned}$$

Proof. By Lemma 3.5(1) one has $|E| \leq N$. By part (2) of the same lemma, every (p, H) -exception already arises as a (p, K) -exception for some $K \in \mathcal{S}_p^\infty$, hence belongs to E .

Fix $H \in \mathcal{S}_p \setminus \{\emptyset\}$, an integer tuple $\overline{m}$, a nonzero integer m , a prime p , and $r \geq 1$, and an integer k such that either 0 or G/H is discrete. Put

$$c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H \in G/H.$$

We first show that

$$c_i \equiv_{p^r} m(b/H) \iff c_j \equiv_{p^r} m(b/H)$$

for all $i, j \in C$. By indiscernibility, either $c_i \equiv_{p^r} c_j$ for all distinct $i, j \in I$, or else $c_i \not\equiv_{p^r} c_j$ for all distinct $i, j \in I$. The claim is trivial in the first case. In the second case, there is at most one $i_0 \in I$ such that $c_{i_0} \equiv_{p^r} m(b/H)$. If this i_0 exists, then i_0 is a (p, H) -exception, so $i_0 \in E$ and the claim follows.

Applying this with $k = 0$ as H varies, we obtain that $\mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_i - mb)$ is constant on C . If the common value is $\emptyset$, the predicate $\overline{m} \cdot \overline{a}_i - mb \equiv_{p^r, \bullet} k_\bullet$ fails on C . Otherwise, by [7, Lemma 2.21(2)], a witness to the predicate is this common value H . If G/H is dense, the predicate fails on C ; if G/H is discrete, the equivalence above proves the constancy of the predicate on C . $\blacksquare$

Theorem 3.7. *Let G be a nontrivial ordered abelian group with finite spines, and suppose that*

$$N = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. Then

$$\text{dp-rk}(G) \leq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Proof. We work in a sufficiently saturated monster model. Let $(\bar{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. Apply Lemma 3.4 and Lemma 3.6. Dividing I by the common order-cut point together with the at most N points of E produces at most $N + 1$ points.

Fix a convex component C of the remainder, and let $i, j \in C$. Let M_i be the subgroup generated by the entries of $\bar{a}_i$ together with b , and define

$$f_{ij} : M_i \longrightarrow M_j$$

by

$$f_{ij}(\bar{m} \cdot \bar{a}_i + mb) = \bar{m} \cdot \bar{a}_j + mb.$$

The case $m = 0$ in the following verifications follows directly from indiscernibility. For $m \neq 0$, Lemma 3.4(1) shows that the sign of every element is preserved. Note that $\bar{m} \cdot \bar{a}_i + mb = \bar{m}' \cdot \bar{a}_i + m'b$ if and only if $(\bar{m} - \bar{m}') \cdot \bar{a}_i + (m - m')b = 0$. This shows that f_{ij} is well-defined and injective. It also follows that f_{ij} is an ordered group embedding. The same lemma, parts (2) and (3), shows that f_{ij} preserves every $\mathfrak{t}_p$ and every predicate $x =_{\bullet} k_{\bullet}$. By Lemma 3.6, it preserves every $\mathfrak{s}_{p^r}$ and every predicate $x \equiv_{p^r, \bullet} k_{\bullet}$. Therefore, Proposition 3.2 implies that f_{ij} is elementary. Since it fixes b and sends $\bar{a}_i$ to $\bar{a}_j$, we obtain

$$\text{tp}(\bar{a}_i/b) = \text{tp}(\bar{a}_j/b).$$

We have verified the hypothesis of Fact 3.1 with $n = N + 1$. Hence $\text{dp-rk}(G) \leq N + 1$, as required. $\blacksquare$

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